

Counting Donuts

Math 252

Donut problems

1. Danielle's Donuts sells 12 kinds of donuts. This week they are offering a special: The Mighty Mixer. You get three donuts of three different kinds for \$1.99. How many different Might Mixer orders are there?
2. Debra has purchased a Mighty Mixer consisting of one sprinkle donut, one jelly-filled donut, and one pumpkin spice donut. She decides to give them away to some of her friends. She has twelve friends.
 - a. How many different ways can she give away the donuts if no one gets more than one?
 - b. How many different ways can she give away the donuts if she allow the possibility of giving more than one to the same person?
3. Don's Donuts sells 3 kinds of donuts: plain, chocolate covered, and glazed. You decide to order a different dozen (12) every week until you have ordered every possible dozen. How many weeks will that take?
4. What are the key features that make these situations different from each other? Can you work out a general formula for each of the situations?
5. Write an additional problem of each type (and work out the answer). Do not use the numbers 3 or 12 in your problems. Do not use donuts.